

Joshua Kim

Curriculum Vitae

Department of Computer Science
University of Texas at Austin

Research Interests

- Computer architecture and hardware-software co-design for datacenter efficiency.
- Acceleration of complex problems using heterogeneous architectures.
- Compiler tools that offload resource management and parallel code from programmers.

Education

- 2025–Present **Ph.D Computer Science**, *University of Texas at Austin*
Advisors: Prof. Jovan Stojkovic, Prof. Christopher J. Rossbach
Topic: Designing Processor Architectures for Datacenter Servers
- 2021–2025 **Bachelor of Science, Computer Science & Mathematical Sciences**, *Carnegie Mellon University, (Dual Degree)*
Thesis: Towards a Scalable Multi-GPU System for Encrypted AI **GPA:** 3.73/4.0

Conference Publications

- C2. Siddharth Jayashankar, **Joshua Kim**, Michael B. Sullivan, Wenting Zheng, and Dimitrios Skarlatos. “Cerium: A Multi-GPU Framework for TB-scale Encrypted LLM Inference”, *To appear In Proceedings of the 32nd Symposium on Operating Systems Principles (SOSP)*, September 2026.
- C1. **Joshua Kim**, Chaojie Zhang, Íñigo Goiri, Christopher J. Rossbach, and Jovan Stojkovic. “PhaseWeave: Phase-aware Execution on Heterogeneous Chiplet Architectures for Datacenters”, *In Proceedings of the 53rd International Symposium on Computer Architecture (ISCA)*, June 2026.

Workshop and Other Publications

- W3. **Joshua Kim**, Chaojie Zhang, Íñigo Goiri, Christopher J. Rossbach, and Jovan Stojkovic. “Hunting for Offload: Automated Discovery of Acceleratable Code in Datacenters”. *In 8th Young Architect Workshop (YArch’26) with ISCA*, June 2026
- W2. **Joshua Kim**, Chaojie Zhang, Íñigo Goiri, Christopher J. Rossbach, and Jovan Stojkovic. “Hunting for Offload: Automated Discovery of Acceleratable Code in Datacenters”. *In Architecture 2.0: Workshop on AI for Computing Systems Design with ASPLOS*, March 2026
- W1. **Joshua Kim**, Chaojie Zhang, Íñigo Goiri, Christopher J. Rossbach, and Jovan Stojkovic. “Lightweight AI for Efficient Resource Management in Heterogeneous-Core Architectures”. *In Architecture 2.0: Workshop on AI for Computing Systems Design with ASPLOS*, March 2026

Scholarships & Awards

- 2026 **Student Travel Grants**, ASPLOS'26, ISCA'26
- 2025 **Allen Newell Award for Excellence in Undergraduate Research**, Carnegie Mellon University
- 2024-2025 **CRA Outstanding Undergraduate Research Awards - Honorable Mention**, Computing Research Association

Research Experience

- Aug 2025 - **Research Assistant, UT Austin**, *Advisors: Prof. Jovan Stojkovic, Prof. Christopher J. Rossbach*, Designing Processor Architectures for Datacenter Servers
Present
- May - Aug 2026 **Intern, Azure System Research, Microsoft Research**, *Advisors: Dr. Chaojie Zhang, Dr. Íñigo Goiri*, Redmond, WA.
- Aug 2024 - **Undergraduate RA, Carnegie Mellon University**, *Advisor: Prof. Dimitrios Skarlatos*, Towards a Scalable Multi-GPU System for Encrypted AI
Aug 2025
- Nov 2023 - **CS Academy, Carnegie Mellon University**, *Advisor: Prof. David Kosbie*
May 2024 Clustering in Plagiarism Detection

Teaching Experience

- Fall 2026 **Graduate TA**, *UT Austin*, CS-395T: Hardware-Software Co-design for Cloud and Datacenters
- Fall 2024 – **Undergrad TA**, *Carnegie Mellon University*, 15-213/513: Introduction to Computer Systems (*Head TA*)
Summer 2025
- Spring 2025 **Undergrad TA**, *Carnegie Mellon University*, 15-418/618: Parallel Computer Architecture
- Fall 2023 – **Undergrad TA**, *Carnegie Mellon University*, 21-259: Calculus in Three Dimensions
Spring 2024

Technologies

Languages: C, C++, CUDA, Python, Java, OCaml, SML, x86/64 Assembly, SQL

Systems/Infrastructure: Kubernetes, Kueue, KubeRay, Ray, Docker

Hardware: CPU, GPU, AWS Trainium

Architecture Simulators: SST, QEMU, Pin

Benchmarking/Profiling: DCPperf, Intel Perfspect, VTune, CMT-CAT, Nvidia-SMI, Nsight Systems, McPAT